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IPTV Kicked off in Shanghai

– *Telecom and Cable TV Operators for the first time in Close Collaboration for Commercial Telecom Service Offerings*

December 8, 2005

Last week was quite a week for IPTV in China: On Monday (Nov.28th) China Telecom announced that it formed an alliance to expand IPTV trial service to more of its territories, and on Tuesday Shanghai began its commercial launch of IPTV services. Both cases involve telecom carriers and cable TV operators, collaborating to offer commercial telecom services for the first time.

China is now caught up in an IPTV reverie with industry analysts and media pundits proclaiming IPTV to be the next great potential telecom market in China. Most telecom carriers and equipment vendors now rank IPTV as a top priority of their development strategy. Figures from CCID, the research unit of the Ministry of Information Industry (MII), show that by the end of this year, China will have about 300,000 IPTV users, growing six times from the number of 50,000 in 2004. CCID estimated that by 2009, the number of IPTV users in China will reach 14.5 million[1].

There are several reasons for the interest in IPTV. First, there is a desire to stimulate the market weary with the protracted wait for pending 3G licenses and to find "killer applications" to reverse the declining trend in ARPUs. Carriers see IPTV as a means to promote the uptake of their broadband services; and equipment vendors stand to boost their sales by providing IPTV equipment.

While convergence has been studied for some time, China's telecom and cable sectors remain exclusive from each other offering distinct services and regulated by two different agencies. However, convergence has become a key goal of China's just released 11th Five-Year Plan (2006-2010), which is scheduled for approval in the next meeting of the National People's Congress next March. Under the plan it is clear that China's State Council is intent on achieving convergence of networks and the services over the next five years. Thus the recent

[1] Source: CCID Consulting, and www.video.com.cn

announcements in Shanghai of cooperation between cable and telecom service providers will be very closely watched precedent for the convergence of the two sectors.

National Rollout

This cooperation has moved this past week to the full commercial phase after three years of trial operation. The service was launched jointly by Shanghai Telecom, China Telecom's Shanghai branch, and Shanghai Media Group (SMG), Shanghai's lead broadcaster. The monthly service charge is RMB 60 (USD 7.4). Adoption of the service in the initial phase is being incentivized through provision of free set-top boxes and a 50% discount fee (RMB 30/month). Press statements from the companies claim that their networks can collectively serve millions of households IPTV services, compared to capacities of no more than 100,000 in commercial trials being held elsewhere. Industry experts believe that this is a substantial step in the national approach to convergence between the telecom carriers and cable TV operators[2].

The day before Shanghai's IPTV commercial launch, China Telecom held a press conference announcing its alliance with Changhong Electronics and SMG to jointly push IPTV services in 17 cities across the country. The cooperation is said to leverage China Telecom's strength in its national broadband network, SMG's strong cable TV subscriber base and content services, as well as Changhong's set-top box and television manufacturing capabilities. IPTV trials have already been launched in many local branches of China Telecom for some time, with large scale commercial launches likely to follow now as part of a nation wide promotion effort.

China Netcom, the other fixed-line telecom operator in China, also started IPTV trial operations through its branches in a few provinces like Heilongjiang. According to a 25th November report of the Xinhua News, China Netcom is close to signing an agreement with SMG to expand its IPTV tests to 21 cities in ten municipalities and provinces across Northern China.

SARFT Regulation Provides Leading Role for SMG

Currently within the IPTV operation, telecom carriers offer the communication pipe for transmitting data flow, while cable TV operators provide the content and the cable TV subscriber base. Last year the State Administration of Radio, Film and Television (SARFT) promulgated "Document 39" which mandates licenses for any service provider seeking to provide any Internet-based video or audio services, either through IP networks or mobile networks and handsets. SMG is so far the only company to have been awarded with such an IPTV license from SARFT. Thus until such time as other licenses are issued, under Document 39, any commercial IPTV offerings need to be offered in cooperation with SMG.

SMG is the largest broadcasting conglomerate in Shanghai. It was formed in 2001 from the

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[2] Source: www.TVoverIP.com, http://www.tvoverip.com.cn/IPTV.asp?column_id=2662

merger of several broadcasters including the People's Radio Station of Shanghai, East Radio Shanghai, Shanghai Television Station and Oriental Television Station. SMG owns 11 analogue TV channels, 90 digital paid cable TV, a full broadcasting Internet TV service, along with 10 analogue and 19 digital radio services[3]. SMG groups together the leading content providers of Shanghai's cable TV industry as part of a state level effort since the late 1990s to "separate TV stations and networks"[4]. SMG remains under the strict control of SARFT.

China's cable TV network is not able to support large-scale data services. Although there are over 140 million cable TV subscribers, only about 0.5% of these connections have been upgraded to handle two-way data services like IPTV.[5] While SMG historically enjoys very close ties with the local cable service provider, SMG can not afford to wait for an IPTV ready cable network upgrade which would take at least three years to complete. For these reasons SMG has turned to the telecom operators to realize their IPTV ambitions.

There have been several factors bringing the telecom operators closer to the cable providers as well. China's major fixed line telecom operators (China Telecom in South China, and Netcom in North China) have been operating their own computer-based IPTV trial services over the past couple of years. However, under SARFT Document 39, they have been prohibited from offering any real-time TV programs or interactive video services. The telecom carriers fully understand on the other hand the huge potential TV-based IPTV. As of H1 of 2005, China's cable TV subscriber base had reached 140 million, almost three times the size of the 53 million broadband telecom subscriber base[6]. Add to this the general preference to watch video on a TV over a PC, plus the fact that there is much room for a yet much larger IPTV subscriber base with over 400 million TV sets in use in China.).

In 1999, the State Council issued Document 75 which stipulated that telecom operators couldn't operate cable TV services and vice versa. This was reinforced that same year with the joint MII/SARFT issued Document 82, a "hongtou wenjian" (red letter headed official circular)[7] entitled "Opinions to Strengthen Cable Network Construction and Management", Document 82 opened a way for the two systems to experiment in convergence around IPTV in Shanghai. Consequently, Shanghai had over five years convergence trials before the IPTV formal launch happened.

Not Yet Triple Play

The IPTV service offered in Shanghai includes direct-broadcasting, interactive TV and

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3 Source: SMG website: www.SMG.sh.cn

4 The "separation of TV stations and Networks" call was an effort to develop the cable networks across the country, as well as to leave the TV stations more rooms to choose program sources. Due to close relations between them, however, recently it has again become national trend to reunite them into one company. In July this year, CCTV acquired China Cable Network, indicating that several years' separation drive has been suspended and reunification is encouraged. Actually before this acquisition, a few similar cases had happened in local levels.

5 Statistics from MII, www.MII.gov.cn

6 Source: CCID Consulting

7 Rules or stipulations released by Chinese government organs to regulate the operations in certain sectors or industries.

information services[8]. Voice services, such as traditional fixed-line telephony and VoIP are still excluded from the offerings. Though IPTV has opened up an initial area of collaboration, telecom carriers and cable TV operators remain cautious about full business integration, making full “triple play” cooperation including voice services unlikely to happen for at least the next few years.

“Definitely no voice services in Shanghai’s IPTV offering, and the service is only open to ADSL subscribers, excluding LAN and cable line subscribers,[9]” Wu Huaxuan, a senior researcher with the Interfax Shanghai branch, told USITO. He visited some outlets of Shanghai Telecom where the IPTV is purchased and interviewed a few senior staff. He called it “double-play” – interactive TV and broadband Internet access. He commented that even for the double-play there remained lots of technical issues to solve, e.g., the quality of the TV programs is not good enough, moreover, the existing ADSL modem does not support IPTV, any ADSL subscribers that apply the service have to pay an extra RMB 150 to get an IPTV-adaptable ADSL modem.

The triple play in China was initially pursued by the cable sector in the late 1990’s. It was strongly opposed by the telecom carriers, fearing it would jeopardize their core voice revenues. In 2000, the Shanghai Oriental Pearl Group (SH, 600832), another cable TV operator in Shanghai, saw its stock price rocket on its “first triple play in China”, a proposal to establish a “Three-in-One network”, offering telephone, internet and television services to cable TV subscribers. To date they have still not realized this dream.

Since the late 1990’s China’s major telecom operators have also began preparing for the triple play, with China Telecom launching large scale upgrading projects for its metropolitan-area networks (MAN) to increase capacity to handle more advanced data services. All layers of its network, including the access network, transmission network and QoS capabilities have been significantly improved.[10] Currently, China Telecom is technically capable to offer triple play in tier 1 cities like Shanghai, Beijing, Guangzhou and Shenzhen. With the networks ready, it has recently placed an emphasis on the terminals, like the recent focus on TV sets. China Telecom signed a strategic agreement with Changhong early this year to cooperate in TV sets production, vowing to make every TV set IPTV ready in the future with built-in Internet access chips.

To protect its market share in the face of growing competition in the telecom Value Added Service (VAS) market, China Telecom CEO Wang Xiaochu has been working to reposition the company as a “modern comprehensive information service provider”. The “triple play” offering is regarded as core to this repositioning strategy. China Telecom is pursuing this strategy by partnering with TV vendors to leverage their strength in producing terminals as well as

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[8] Source: press release issued by Shanghai Telecom and SMG, tech.sina.com.cn

[9] Shanghai, like most other regions in China, has three Internet access means: ADSL, fiber (LAN), and cable line. ADSL service is solely operated by Shanghai Telecom, with the number of its subscribers constituting about 80% of the total broadband subscribers of the city.

[10] Source: www.chinatelecom.com.cn

exploiting their sales channels to residential consumers.

Standards Issue

To date, China has not decided on which IPTV standard to adopt. According to the China Academy of Telecom Research, a research unit of the MII, the IPTV standard will address seven sub-sectors of the IPTV industry: set-top boxes, business platform interfaces, operation platform interfaces, access equipment support, business demand, video coding and decoding system and the overall IPTV framework. The new standard will be a complex one with technologies in the seven sectors specified and consolidated. Currently competition is most active for the video coding and decoding system, including MPEG-4, H.264, and China's homegrown Audio Visual Standard (AVS).

The two existing dominant video coding standards for IPTV, MPEG-4 and H.264, have both been adopted by Shanghai's IPTV networks in advance of the release of Chinese final IPTV coding standard (see USITO Weekly for Nov 21, 2005). According to a report in the Daily Economic News, H.264 was deployed in Shanghai's Pudong New District, and the MPEG-4 standard in the Minhang District. An official with Shanghai Telecom has stated that a unified standard will be decided in the second half of 2006. The alliance for the homegrown Audio Visual Standards (AVS) is also striving to make AVS adopted as China's final official standard. In order to incentivize the local industry, Zhou Pengyuan, deputy secretary of AVS alliance, has stated that China's IPTV service providers will not be charged for using AVS standard.

According to an official with SARFT, the final decision for the coding and decoding standard will also depend on which business will become more established sooner in China. MPEG-4 could be adopted first if there is initial market traction in TV based video services like VOD, rather than the Interactive value-added businesses which would favor H.264. The prospects for the domestic AVS standard are not bright since there have to date still been no successful large scale trials, leaving big question marks around its technical and commercial readiness.